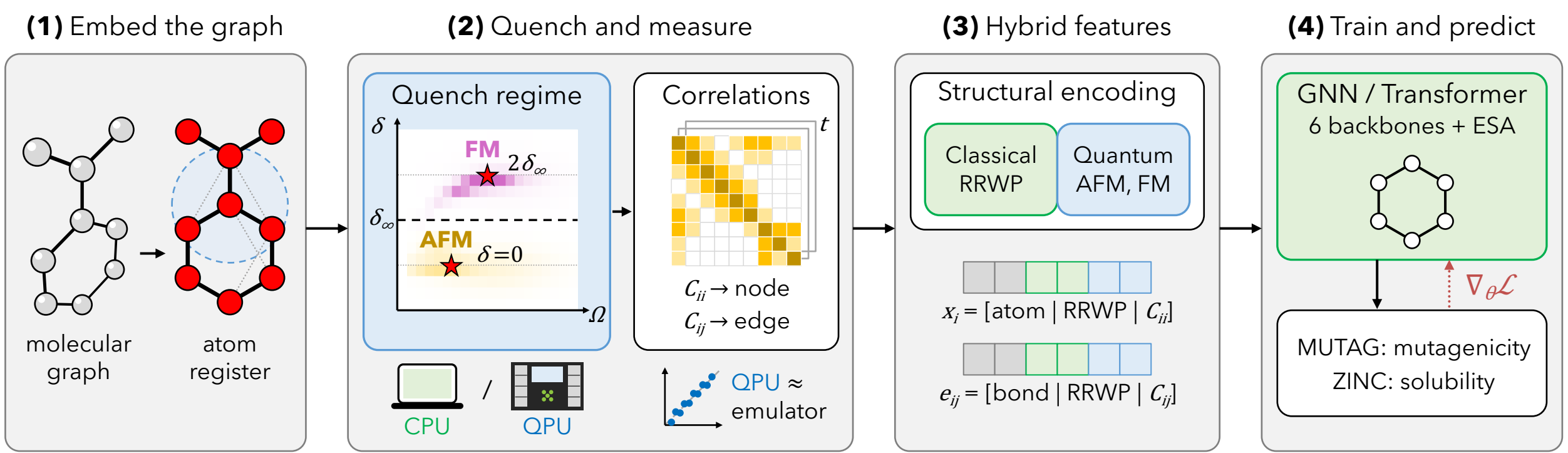

(1) Embed the graph
molecular graph
atom register
(2) Quench and measure
Quench regime
δ
δ∞
FM
2δ∞
AFM
δ=0
Ω
Correlations
t
C_ii → node
C_ij → edge
CPU
/
QPU
QPU ≈ emulator
(3) Hybrid features
Structural encoding
Classical RRWP
Quantum AFM, FM
x_i = [atom | RRWP | C_ii]
e_ij = [bond | RRWP | C_ij]
(4) Train and predict
GNN / Transformer
6 backbones + ESA
∇θℒ
MUTAG: mutagenicity
ZINC: solubility